LESSON 5.3
MAKING POPULATION PREDICTIONS - PART 1

LESSON INTRODUCTION

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

LEARNING TARGETS

- I can use proportional relationships to make predictions about a population.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.

ESSENTIAL QUESTIONS

- How can data from a sample be used to make predictions about a population?
- How can fractions and percentages be used to make population predictions?

LESSON NARRATIVE

Given data in multiple forms and using some new situations, students use the data to calculate percentages of the population that fit the given situations. They are introduced to three different strategies to use for making predictions about the populations using calculated percentages from sample data. The lesson provides opportunities for them to understand and analyze each strategy as they determine which they prefer or which works best for different situations.

COMPONENT SUMMARY

5.3.1 WARM-UP *(~5 MIN.)*

Notice and Wonder about a restaurant's prices

5.3.2 GUIDED INSTRUCTION *(10-15 MIN.)*

Learn to use equivalent fractions, scale factor, and percentages to make predictions about a population from a sample

5.3.3 EXPLORATION *(10 MIN.)*

Explore how to use probability to make predictions

5.3.4 GUIDED PRACTICE *(10-15 MIN.)*

Synthesize learning to practice problems with teacher's support

5.3.5 WRAP-UP *(~5 MIN.)*

Reflect on learning and understanding of today's learning target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may get confused because different strategies for making population predictions may result in slightly different answers.
- Students may also get confused by the way the word “estimate” is used in this lesson. They need to understand that the prediction is an estimate because it can’t be known for certain, but they should still be finding exact values based on the predicted percentages from the sample data.

THINGS TO CONSIDER

- At this time, the focus in problem-solving for these types of questions is NOT on using proportions. Unit 6 will focus on setting up and solving proportions. The focus in this unit is on using strategies such as equivalent fractions, scale factor, and percentages.
- In problems that require several steps, students should not round until the final step. Rounding midstep could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

The 5.3.1 Warm-Up allows students to explore, wonder, and discuss their mathematical ideas in regards to a restaurant menu. It can open up the idea of analyzing differences and discovering if changes are consistent (proportional).

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

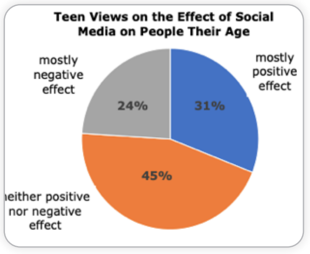
MA.K12.MTR.6.1 Reasonableness

5.3.3 Exploration gives students an opportunity to make predictions and reflect on the reasonableness of their predictions as well as those made by other students using different strategies.

DIFFERENTIATE INSTRUCTION

The 5.3.3 Exploration allows the teacher the opportunity to make informal observations to inform groupings during 5.3.4 Guided Practice. Consider whether students are ready to work independently or with partners and which partnerships would best support all students. It may also be appropriate to pull a small group during the 5.3.4 Guided Practice based on observations during the 5.3.3 Exploration.

Additional differentiation opportunities are denoted throughout the lesson guide.



5.3.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Highlight student responses that mention the following:

- The price difference between adding chips or fries is not consistent and varies from a $0.45 - $0.95 difference.
- The price to add cheese ranges from $0.30 - $0.80.
- Is there a pattern to the differences?
- Is the amount of fries different when the cost is higher?

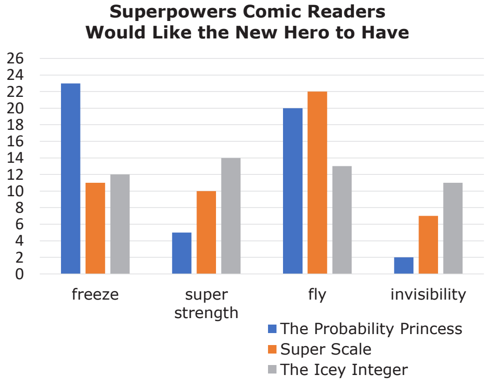
5.3.1 Warm-Up: Notice and Wonder

Burgers	Chips	Fries
Deluxe Burger	$6.90	$7.70
Deluxe Cheese	$7.70	$8.45
Bacon Burger	$7.95	$8.65
Bacon Cheese	$8.25	$8.95
Double Burger	$8.35	$9.30
Double Cheese	$9.25	$9.95
Double Bacon Cheese	$9.50	$10.25
Cheese Garden Burger	$8.25	$8.85
Mushroom Swiss	$8.25	$8.95
Jalapeno Pepper Jack	$8.25	$8.95
BBQ Western Bacon	$9.20	$9.85
Pastrami Burger	$9.20	$9.85
Power Pac	$10.45	$11.05
Cheese Power Pac	$11.25	$11.70

1. What are two things you notice about this menu?
   Answers will vary.
2. What is one thing you wonder?
   Answers will vary.
3. What is a suggestion you could give to the owners about their pricing?
   Answers will vary.

5.3.2 Guided Instruction: Predicting How Many

Results of a random sample can be used to make predictions. The results from a sample of 50 readers of three different comic books are shown.



Suppose the author of *The Probability Princess,* Ashley Grant, is trying to make decisions about the future of the series. She really wants the new hero to have the ability to fly, but she knows there is a larger percentage of readers who want the new hero to have freeze power. As long as the new series can sell at least 3,600 issues in the next five years, Ashley will be able to stay in business.

1. She determines 50 readers of *The Probability Princess* responded to the survey. She then finds the percentage of the 50 readers surveyed who want the new hero to have the ability to fly.

$$\frac{\boxed{20}}{50} = \frac{40}{100} = \boxed{0.4} = 40\%$$

Over the past five years, *The Probability Princess* has had 10,000 loyal readers. Using three different strategies, Ashley uses the data to predict the number of those readers who want to see the new hero have the ability to fly. In each method, she predicts that 4,000 of the 10,000 readers want this.

Equivalent Fraction	Scale Factor	Percent
The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Write equivalent fractions until the denominator shows the population total (in this case 10,000). $\frac{20}{50} = \frac{40}{100} =$ $\frac{400}{1{,}000} = \frac{4{,}000}{10{,}000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Divide the population by the sample to find the scale factor needed. Then, multiply both the numerator and denominator by this scale factor to write an equivalent fraction. $10{,}000 \div 50 = 200.$ $\frac{20}{50} \times \frac{200}{200} = \frac{4{,}000}{10{,}000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Find the percent of the sample for the given condition. Then multiply the population by the percent. $20 \div 50 =$ 0.4 or 40% $10{,}000 \times 0.4 = 4{,}000$ OR $40\% = \frac{40}{100} = \frac{2}{5}$ $10{,}000 \times \frac{2}{5} = 4{,}000$

5.3.2 GUIDED INSTRUCTION: PREDICTING HOW MANY

TEACHER MOVES

The questions in this section are designed to show students how population predictions from survey data can be used to make business decisions. Where else might this be a useful calculation?

TEACHER MOVES

It is important that students understand how each of these strategies work. The discussion questions are meant to provide scaffolding.

5.3.2 GUIDED INSTRUCTION: PREDICTING HOW MANY (CONTINUED)

ENRICHMENT

Students may struggle to answer Question 5. Consider asking the following questions to further scaffold and guide student thinking:

- How many issues must Ashley sell in the next five years to stay in business? (At least 3,600.)
- How many of her loyal readers does she predict will still buy the series if the new hero has the ability to fly? (At least 4,000 of the 10,000 loyal readers will still buy the series if the new hero has the ability to fly.)
- • How does she determine if she can keep enough readers to keep in business? (Since 40% of the readers surveyed want the hero to fly, she figures 40% of the loyal readers will continue to purchase the series, which is 4,000 people and should translate into at least 4,000 issues being sold over the next five years. Since 4,000 is greater than 3,600, or the number of copies she needs to sell to stay in business, Ashley concludes she can give the new hero the ability to fly.)

2. Explain whether one of Ashley's strategies seems to be better than the others for solving this particular problem.
   Answers will vary. All three strategies work equally well. If I were wanting to solve this problem mentally, equivalent fractions would be the easiest way for me to do so.

3. What is the benefit of solving a problem using two different strategies?
   Answers will vary. When you use multiple strategies to solve your problem you are able to check your work. When you use multiple strategies you can understand the problem in a different way.

4. Next, Ashley finds the percent of the 10,000 issues they need to sell in the next five years. Complete the work shown below.

$$\frac{3{,}600}{10{,}000} = \frac{\boxed{36}}{100} = 0.36 = \boxed{36}\%$$

Ashley estimates that if the new hero has the ability to fly, at least 4,000 readers will continue to read and purchase *The Probability Princess*. She estimates that this will lead to each of those 4,000 readers purchasing at least one comic. She also thinks it's likely that readers who did not want the new hero to have the ability to fly may still buy issues in the next five years. Because 4,000 is greater than the 3,600 issues they need to sell to stay in business, Ashley decides to give the next hero the ability to fly.

5. Explain why you agree or disagree with Ashley's decision.
   Answers will vary. I agree with Ashley's decision. She used proportional reasoning – scale factor and equivalent fractions to prove her thinking. She knows that the 4,000 minimum readers is larger than the amount they need to break even.

5.3.3 Exploration: Reaction Times

The track coach at Yulee High School in Nassau County, FL needs a student whose reaction time is less than 0.4 seconds to help out with timers at track meets. All the twelfth graders in the school measured their reaction times.

A random sample of 32 students' reaction times has been recorded in the table.

Reaction Times of Less Than 0.4 Seconds	Reaction Times of 0.4 Seconds or More
22	10

1. What fraction of the students in the random sample have a reaction time less than 0.4 seconds?
   $\frac{22}{32} = \frac{11}{16}$

2. Explain or show how to estimate the probability that a randomly selected twelfth grader from this school has a reaction time of less than 0.4 seconds.
   A random sample of twelfth graders would have a $\frac{22}{32} = \frac{11}{16} = 0.6875 = 68.75\%$ *chance of a reaction time of less than* 0.4 *seconds.*

3. There are 125 twelfth graders at this school. How many of the twelfth graders are expected to have a reaction time of less than 0.4 seconds?
   To find this answer we would use scale factor.
   $125 \times \frac{22}{32} = 85.9375$. *This can be rounded to* 86 *students who are likely to have reaction times less than* 0.4 *seconds.*

4. Suppose another group in your class comes up with a different estimate than yours for the previous question. What is another estimate that would be **reasonable**?
   Another reasonable estimate could be 87.5 *if they rounded the percentage of twelfth graders with a reaction time of less than* 0.4 *seconds to the nearest tenth.*

5. Explain what kind of an estimate would be considered **unreasonable.**
   An estimate of 100 *would be unreasonable. In that instance, rounding to the nearest hundred actually significantly changes the percentage of adjustment. An estimate of* 100 *would be equivalent to about* 8 *tenths of the number of total people rather than* 7 *tenths.*

5.3.3 EXPLORATION: REACTION TIMES

TEACHER MOVES

This question is scaffolded to have students discuss their thinking and strategies as they solve. Students may use different strategies to answer this question.

DIFFERENTIATE INSTRUCTION

Consider using this time to make informal observations to inform potential groupings for the next component, 5.3.4 Guided Practice.

ENRICHMENT

Students often struggle with estimation. Consider asking guiding questions to further scaffold student thinking:

- How can we determine if an answer is reasonable or not?
- What might an unreasonable answer be?
- How can you explain differences in answers?

5.3.4 GUIDED PRACTICE: MAKING POPULATION PREDICTIONS

TEACHER MOVES

Either before or after this component, consider having students brainstorm which strategies they most prefer and when they think each strategy is most useful.

DIFFERENTIATE INSTRUCTION

For students who finish early, create an advertisement using this data. Who would need this data? How could they use it to persuade an audience?

5.3.4 Guided Practice: Making Population Predictions

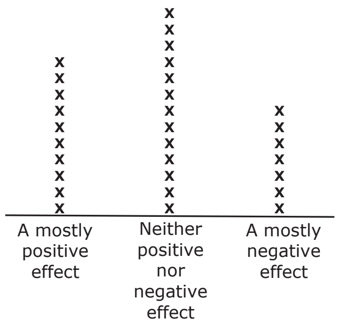
1. Taylor attends Murdock Middle School in Port Charlotte, Florida. She wonders how many students at her school would dye their hair blue if their parents would let them. She surveyed a random sample of 10 students at her school, and 2 of them said they would. Kiran didn't think Taylor's estimate was very accurate, so he surveyed a random sample of 100 students, and 17 of them said they would.

   a. Complete the table below to compare the accuracy of Taylor and Kiran's random samples. Be sure to show your work or explain your thinking.

	Taylor	**Kiran**
<u>Fraction</u> of students surveyed who would dye their hair blue	$\frac{2}{10} = \frac{1}{5}$	$\frac{17}{100}$
<u>Percentage</u> of students surveyed who would dye their hair blue	$\frac{2}{10} = \frac{20}{100}$ $= 20\%$	$\frac{17}{100} = 17\%$
<u>Predicted number</u> of the 500 students in the school who would dye their hair blue	$\frac{20}{100} = \frac{100}{500} =$ 100 *students*	$\frac{17}{100} = \frac{85}{500} =$ 85 *students*

   b. Explain whose random sample likely produced the most accurate estimate.
   Answers will vary. Kiran's sample included more students so that is a more accurate estimate.

2. Ms. Jeffrey teaches seventh grade in Miami, Florida. She noticed many of her students talked about social media and wondered about what her students considered to be the impact of all their online activity. She conducted a random survey of 30 seventh-grade students asking them how they describe the effect of social media use on people their own age. The results are shown in the line plot below.



According to a 2018 Pew Research Center study of American teenagers, 24% of teen respondents said they believed social media has a mostly negative effect on people their own age.

   a. Explain how this compares to the percentage of Ms. Jeffrey's seventh-grade students who described social media as having a mostly negative effect.
   In Ms. Jeffrey's class, there were seven out of thirty students who said they felt social media had a mostly negative effect. $\frac{7}{30} = 0.23 = 23\%$. *This is about equal to what the study had found.*

There were about 1,546,397 teenagers living in the state of Florida in 2018.

b. Predict the number of teenagers in the state of Florida who considered the effect of social media as mostly positive based on the data Ms. Jeffrey collected.
Answers will vary. The number of students in Ms. Jeffrey's class who considered the report mostly positive were $\frac{10}{30} = \frac{1}{3}$. $\frac{1}{3}$ *of* 1,546,397 *is approximately* 515,466 *teenagers. Using these numbers, I could also approximate that about* $\frac{1}{3}$ *of* 1.5 *million is approximately* 500,000 *teenagers.*

c. Predict the number of teenagers in the state of Florida who considered the effect of social media as neither positive nor negative based on the data Ms. Jeffrey collected.
The number of students in Ms. Jeffrey's class who considered the report mostly positive were $\frac{13}{30} = 0.433 = 43.3\%$. $0.433 \times 1{,}546{,}397 = 669{,}589.90$. *I would predict about* 669,590 *teenagers responded this way.*

d. Explain at least two factors that could account for differences in these estimates from the actual number of students who would have described the effect of social media in these ways.
Answers will vary. See below.

Why are they different?
1. *I chose not to round* 43.3%, *but someone may have rounded it to* 40% *or* 45%.
2. *I rounded to the number of one and a half million before dividing by* 3. *Other people may have wanted to directly find the answer instead of estimating.*

5.3.4 GUIDED PRACTICE: MAKING POPULATION PREDICTIONS (CONTINUED)

ENRICHMENT

Students may struggle to extend the data to predict the numbers for the entire state of Florida.

This question provides opportunities for students to consider the ways rounding can impact estimates.

In problems that require several steps, students should not round until the final step. Rounding midstep could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.

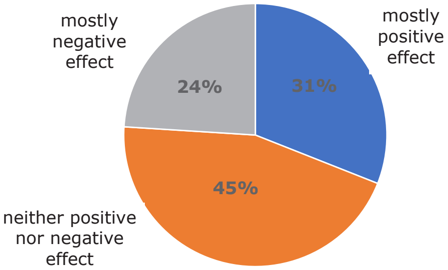
5.3.4 GUIDED PRACTICE: MAKING POPULATION PREDICTIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students are given a visual representation of the data, which lends itself not only to understanding the data, but also to understanding the probability of each event.

The results of the 2018 Pew Research Study are shown in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



e. The percentage of teens who describe social media as having a positive effect on people their age is ○ less than ○ equal to ○ greater than the percent of students who responded the same in Ms. Jeffrey's survey.

Write an explanation of why this may have happened.

Answers will vary. In Ms. Jeffrey's class, $\frac{1}{3}$ of the students rated social media as positive. In the real study it was 31%. If we had used the more exact measure of 31% we would have gotten a more accurate answer.

5.3.5 Wrap-Up: Growth Mindset

Complete the following sentences based on today's learning target:

I can use proportional relationships to make predictions about a population.

1. A barrier to my learning today was …
   Answers will vary. I struggled to remember my multiples and factors so finding the equivalent fractions and scale factors was challenging. I struggled with knowing how much to round when estimating.

2. I will try to overcome this by …
   Answers will vary. I can practice multiplication tables and factor trees. I can practice reasonable estimates by using a number line to assess how far my guess is from the original number.

5.3.5 WRAP-UP: GROWTH MINDSET

DIFFERENTIATE INSTRUCTION

Consider prompting students to use Mark-Up Text to connect their responses in questions 1 and 2 to their learning throughout the lesson. Students can draw a symbol or emoji to represent "barriers," and mark questions with that symbol, providing even more information for their progress from the lesson.

